

EXTERNAL MERGE SORT SIMULATION

ABSTRACT

In contemporary database systems, dealing with large datasets is highly important, especially when the size of the data exceeds the size of the main memory. Conventional in-memory sorting algorithms, such as QuickSort or HeapSort, are not useful in these situations because they require all the data to be held in memory. Instead, database management systems leverage external sorting algorithms, such as External Merge Sort, to sort large amounts of data that is stored on disk. External Merge Sort is characterized as having two primary phases: the run generation phase and the merge phase. In the run generation phase, the large dataset is partitioned into smaller, more compact chunks that can fit into the memory assigned for that operation. Each chunk of the dataset is sorted using an in-memory sorting algorithm and then rewritten to disk as a sorted run. The merge phase then occurs by merging the sorted runs iteratively using a multi-way merge process, which fetches the smallest element in the dataset from k runs at a time, writing it to the final output file. This process is repeated until we arrive at a single sorted dataset.

The main purpose of the project is to mimic the External Merge Sort to demonstrate how it works and its efficiency. It will serve to help visualize how large datasets would process in a disk-based system while showcasing important aspects of this process such as managing memory, performing disk I/O for sorting, and multi-way merging in order to systematically reduce computational overhead. Not only does this project serve an educational purpose, but it also shows how External Merge Sort applies to database query processing; often when a database query requires sorted data in order to perform operations such as ORDER BY, GROUP BY, or even join operations with sorted values. The simulation improves user experience by allowing observation of the number of sorted runs generated, the number of passes in the merge phase, and overall performance of the merging processes - all of which lead to useful considerations in optimizing database operations for large-scale data.

The aim of this project is to connect theoretical knowledge to practice, facilitating a better grasp of the complexities of sorting large data in an authentic database setting. It also serves as a foundation for examining other more advanced topics related to databases, such as query optimization, disk-based algorithms, and performance tuning. By the end of this simulation, one will have a practical means to appreciate how database systems manage and execute data management tasks when the entire dataset cannot fit in main memory, thus demonstrating External Merge Sort as an important concept in the study of Advanced Databases.